Spreadsheet Formulas

14

Spreadsheets: It can be tedious and take up a lot of time to perform mathematical functions on large sets of data. A spreadsheet is a computer application that is used to store, analyse, and organise data in rows and columns.

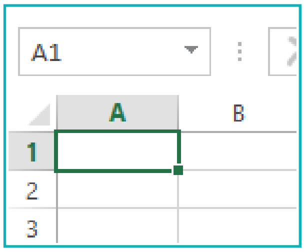
To quickly perform accurate calculations on large numerical data sets, we can use **Spreadsheet formulas.**

Recall:

***Data** is a collection of information like images, words, and numbers. Data can be of different types like numerical information or text description.*

Cell

Each box in a spreadsheet is called a cell. Every cell has a unique name, called a 'cell reference'. It Is a combination of its row number and column heading.



For example:

The first cell in a spreadsheet is named A1. The 'A' stands for the column heading and the '1' stands for the row number.

Cells can contain three types of information:

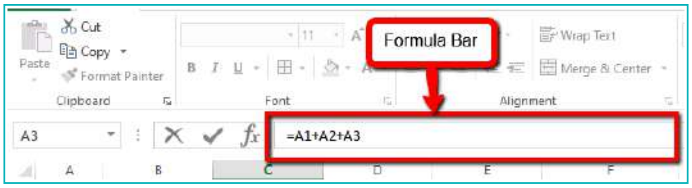
- Numbers
- Text (words)
- Formula

Formulas in a Spreadsheet

- Formulas instruct the computer to perform specific math operations. These cells contain numbers.
- A formula always begins with an equal to (=) sign.
- The formula takes data from selected cells in the spreadsheet and processes it. It performs the specified functions on the data to calculate its value.
- The result is reflected in the cell which contains the formula.
- The formula written in any cell can be viewed in the Formula Bar.

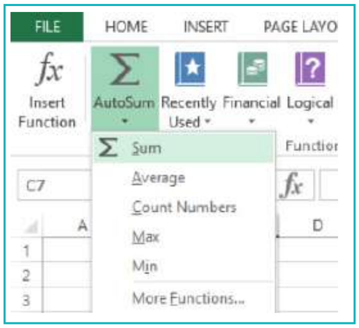
For example:

=A1+A2+A3, which finds the sum of the range of values from cell A1 to cell A3.



Functions

- Excel contains predefined formulas which are inbuilt into its software.
- Functions make it easy to apply formulas to numbers in an Excel sheet.



For example:

=SUM(A1:A3). This function finds the sums of the values from cells A1 to A3.

Examples of Functions

- SUM()
- AVERAGE()
- COLUMN()
- COUNT()

Exercise

Fill in the blanks

1. The formula written in a cell can be viewed in the _________________.
2. A formula always starts with the _________________ sign.
3. Every cell has a unique name which is called a _________________.

State whether True or False

4. Functions make it easy to apply formulas to numbers in Excel.

▲ ______________________

5. Cells can contain only two types of information.

▲ ______________________

6. The SUM() function finds the sum of all values from the given cell reference range.

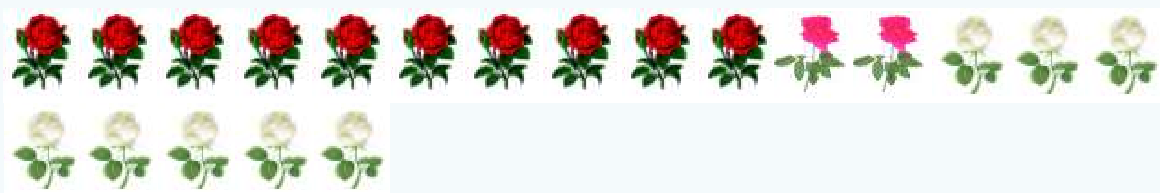
▲ ______________________

Activity

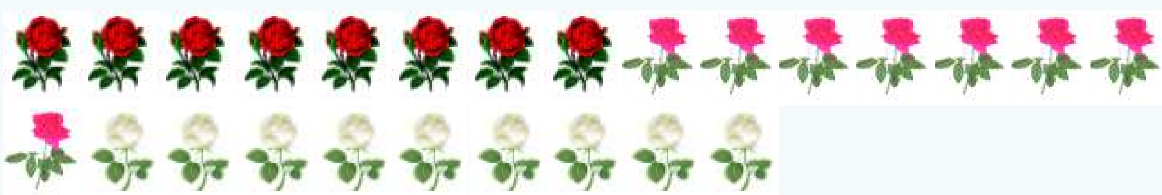
In the data given below, each student has a set of roses of different colours. It takes 5 roses to make a bouquet. We will calculate how many roses each student has and how many bouquets can be created.

Priyanka

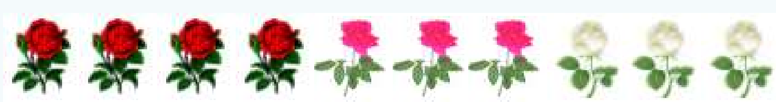
Rajeev



Annsh



Navneet



Mayuri

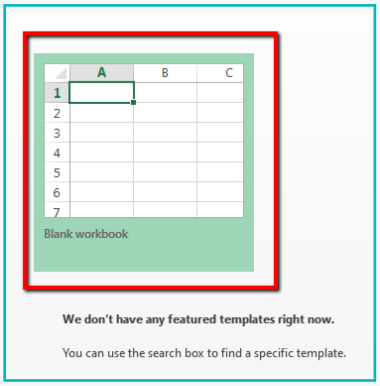
To calculate how many bouquets can be created

1. We will calculate how many flowers each student has.
2. Add the number of flowers each student has, to find out the total number of flowers.
   i.e. Total Flowers = Priyanka + Rajeev + Annsh + Navneet + Mayuri
3. To calculate the total bouquets that can be created: Divide the total number of flowers by 5 (since 1 bouquet contains 5 flowers).
4. After calculating the sum, even if the number of red roses is changed for any student, we will have to calculate the total number of flowers again and then divide it by 5 to calculate the number of bouquets.

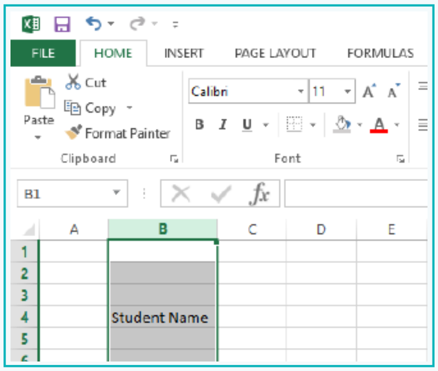
When we create a spreadsheet to calculate data, it saves us time.

Activity

How do we create a Spreadsheet?



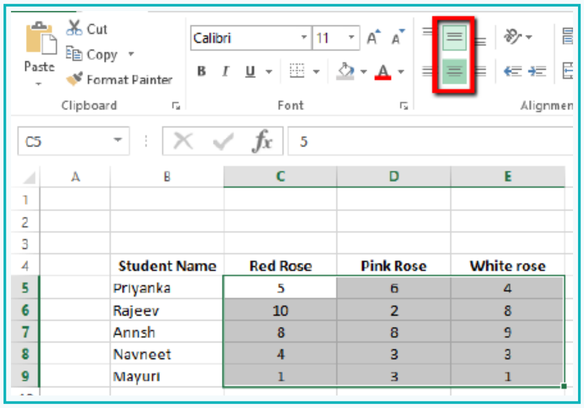
1. Launch the Excel software and click on **Blank workbook.** This will create a new spreadsheet.



2. Columns are usually used for different types of data. In this case, the types of data are: Student Name, Red Rose, White Rose, Pink Rose, and Total Roses.

a. Add Student Name to cell B4. Resize column B to make the text fully visible in that column.

b. Similarly, enter Red Rose, Pink Rose, White Rose, and Total Roses in cells C4, D4, E4 and F4 respectively.



3. Rows are used for entries. In this case, the entries are student names and the quantities of each type of rose. Enter the names of students and the number of roses as per the categories. Centre align the data under the rose categories.

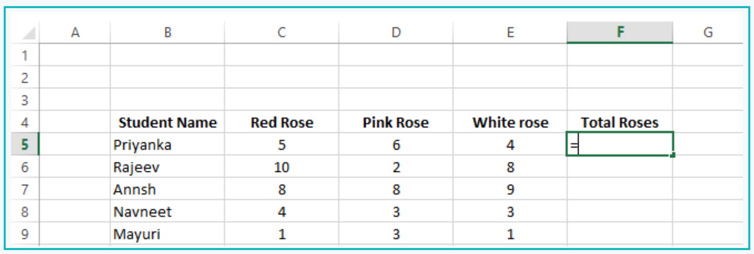
Note:
Centre align all the text and make it bold. This way it is clear and easily visible.

Activity

4 To calculate the total number of roses, there are two methods we can use:

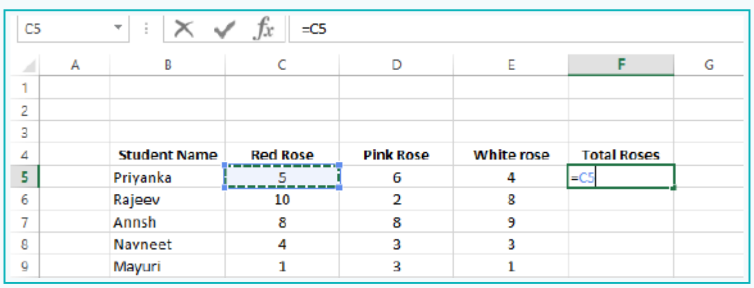
a. Use a formula

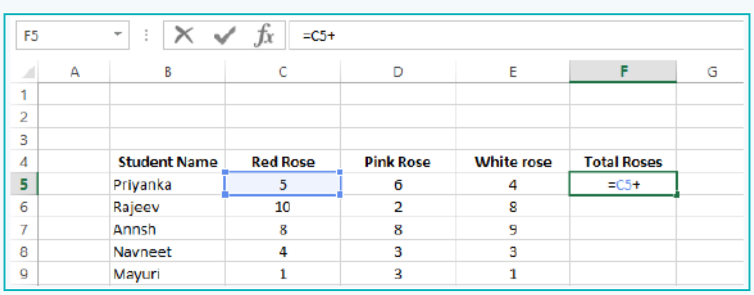
i. Select the cell to enter the formula, under total roses, and type '='.

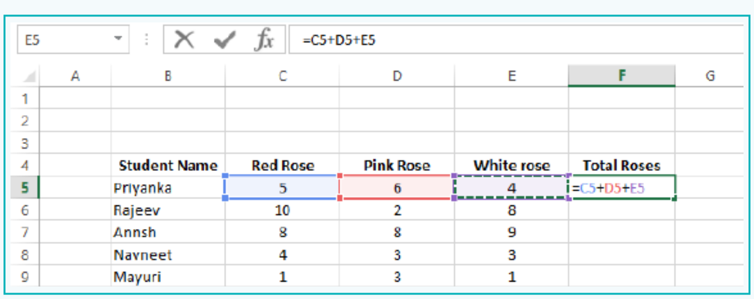


ii. Select the cells whose values are to be added.

For example: In this case, select cells C5, D5, and E5 as shown and press Enter.

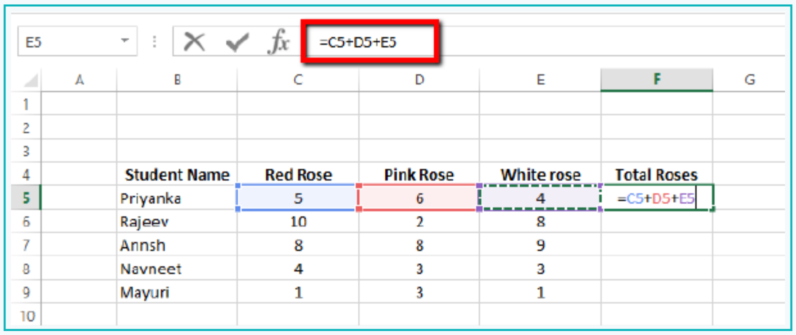






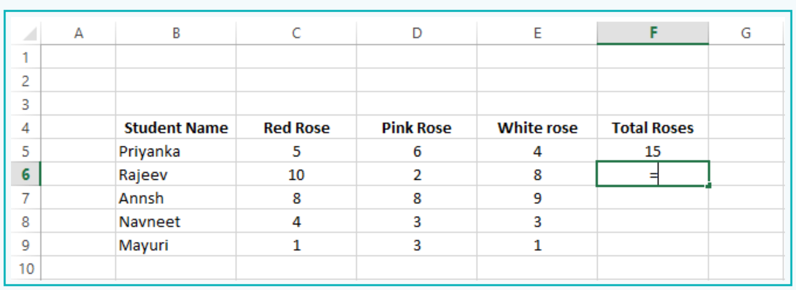
Activity

iii. We can view the formula in any cell in the Formula Bar.



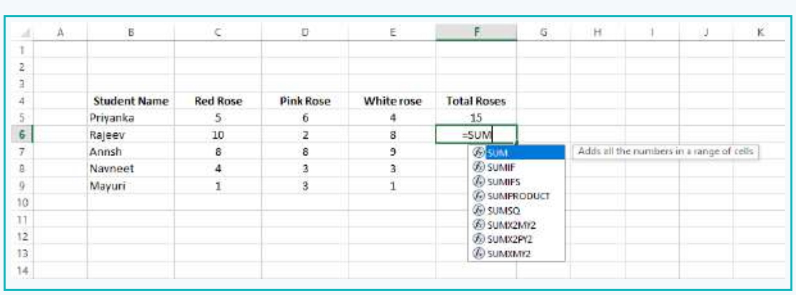
b. Use a function for the formula.

i. Select the cell to enter the function in and type **'='.**



ii. Enter the function of the operation we want to perform.

For example: Here, we want to perform the function SUM on the cells C6, D6, and E6 as shown below.

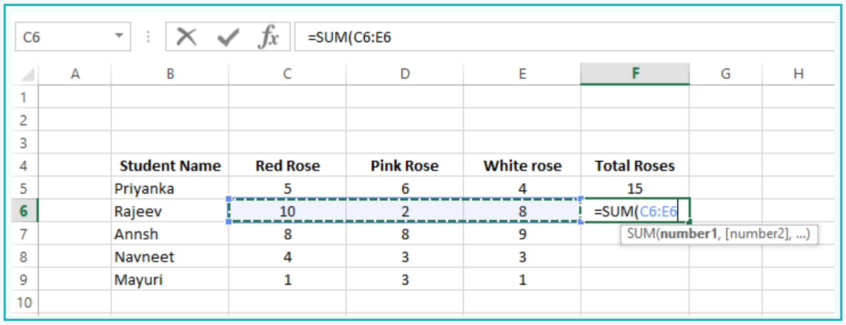


iii. Enter '(' after the SUM function.

Activity

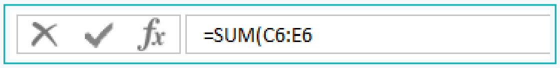
iv. Drag and select the cells to be added.

For example: Drag and select cells C6 to E6 and press Enter.

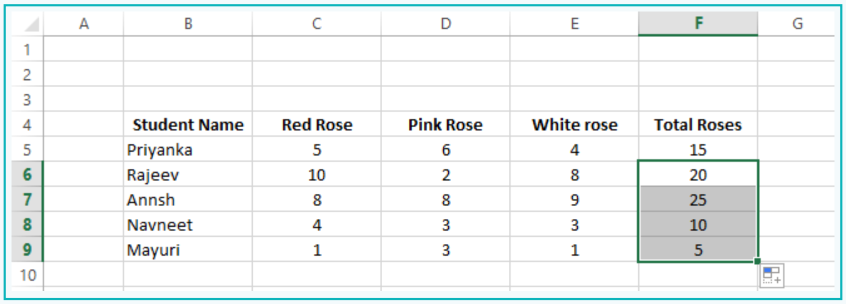


C6 | =SUM(C6:E6

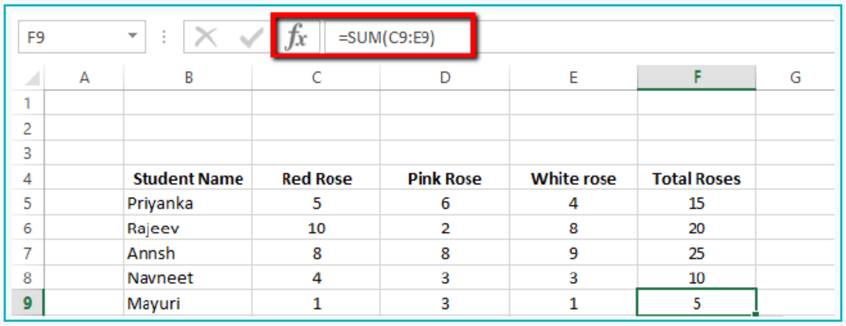
	A	B	C	D	E	F	G	H
1								
2								
3								
4		Student Name	Red Rose	Pink Rose	White rose	Total Roses		
5		Priyanka	5	6	4	15		
6		Rajeev	10	2	8	=SUM(C6:E6		
7		Annsh	8	8	9	SUM(number1, [number2], ...)		
8		Navneet	4	3	3			
9		Mayuri	1	3	1			
10								



5 To calculate the total number of roses for all other students, we don't have to enter the formula separately for each cell. The formula can be dragged from the cell where it has already been applied.



	A	B	C	D	E	F	G
1							
2							
3							
4		Student Name	Red Rose	Pink Rose	White rose	Total Roses	
5		Priyanka	5	6	4	15	
6		Rajeev	10	2	8	20	
7		Annsh	8	8	9	25	
8		Navneet	4	3	3	10	
9		Mayuri	1	3	1	5	
10							



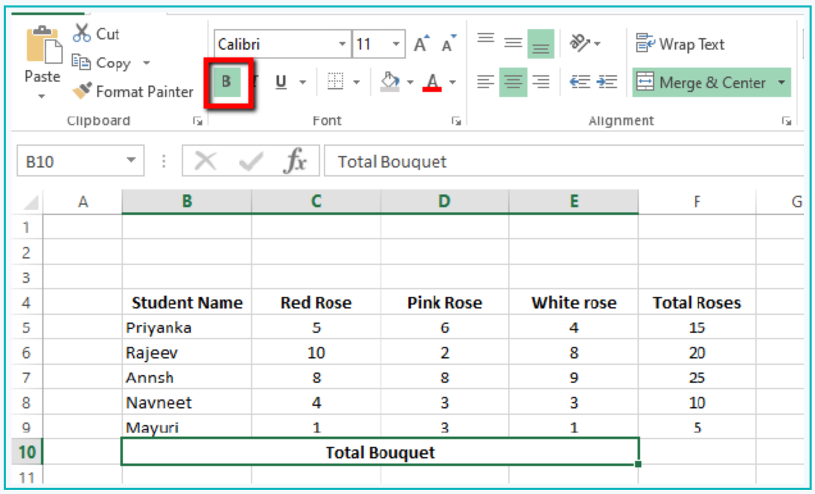
F9 | =SUM(C9:E9)

	A	B	C	D	E	F	G
1							
2							
3							
4		Student Name	Red Rose	Pink Rose	White rose	Total Roses	
5		Priyanka	5	6	4	15	
6		Rajeev	10	2	8	20	
7		Annsh	8	8	9	25	
8		Navneet	4	3	3	10	
9		Mayuri	1	3	1	5	

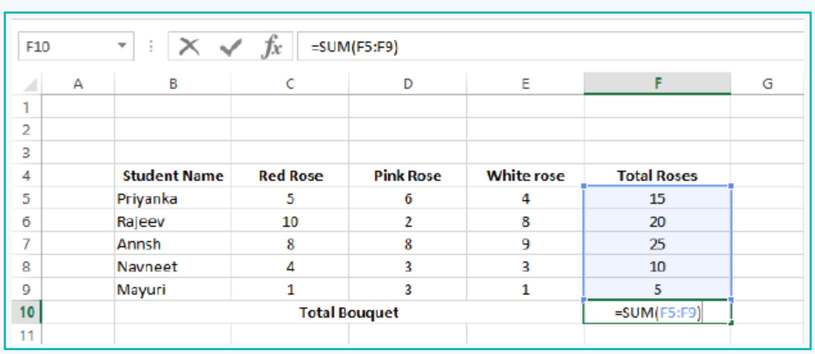
Activity

6 Now, we need to calculate total bouquets that can be created.

a. Select cells from **B10 to E10** and click on **Merge & Centre.** Type **Total Bouquets** in Bold as shown below.

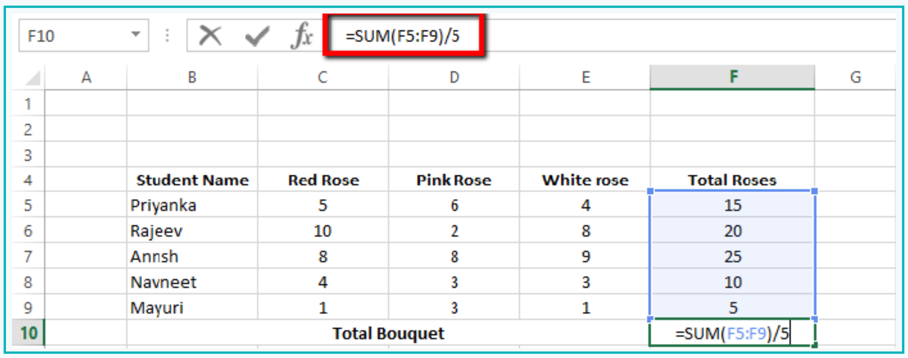


b. In cell **F10**, use the **SUM()** function and select cells **F5 to F9.**



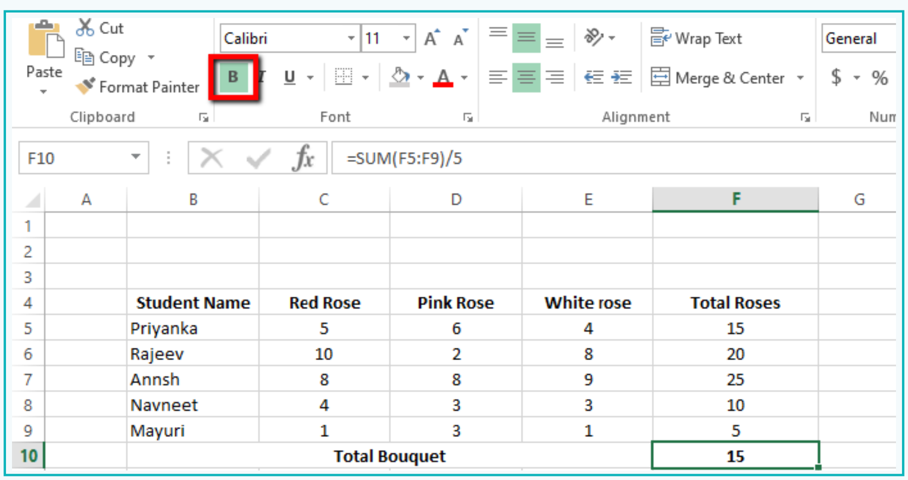
Activity

c. To calculate the total number of bouquets, divide the total number of roses by 5 (5 roses are required to make one bouquet).



	A	B	C	D	E	F	G
1							
2							
3							
4		Student Name	Red Rose	Pink Rose	White rose	Total Roses	
5		Priyanka	5	6	4	15	
6		Rajeev	10	2	8	20	
7		Annsh	8	8	9	25	
8		Navneet	4	3	3	10	
9		Mayuri	1	3	1	5	
10			Total Bouquet			=SUM(F5:F9)/5	

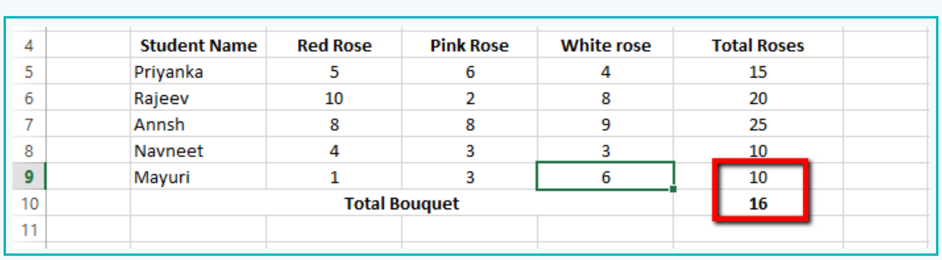
7 Highlight the total number of bouquets and change the font style of cell F10 to Bold.



	A	B	C	D	E	F	G
1							
2							
3							
4		Student Name	Red Rose	Pink Rose	White rose	Total Roses	
5		Priyanka	5	6	4	15	
6		Rajeev	10	2	8	20	
7		Annsh	8	8	9	25	
8		Navneet	4	3	3	10	
9		Mayuri	1	3	1	5	
10			Total Bouquet			**15**	

8 Even if the data changes, the formula remains constant. It will use the cell references to automatically calculate the total number of roses and bouquets.

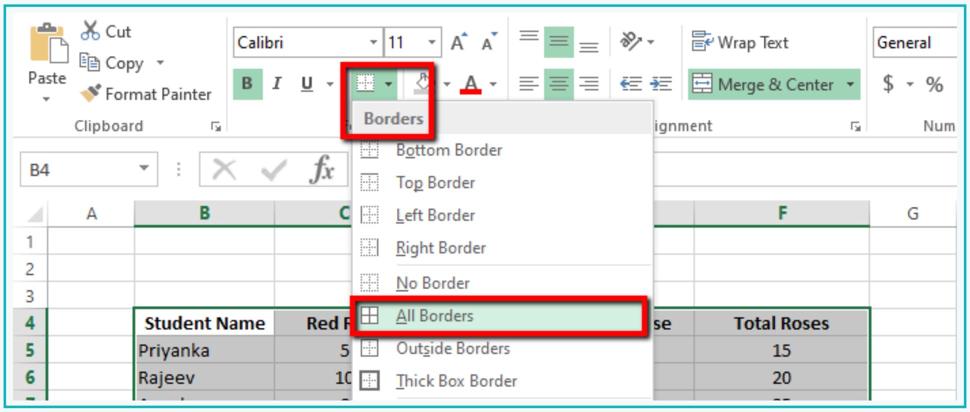
For example: If we change the white roses that Mayuri has, to 6, the total number of bouquets will automatically change to 16.



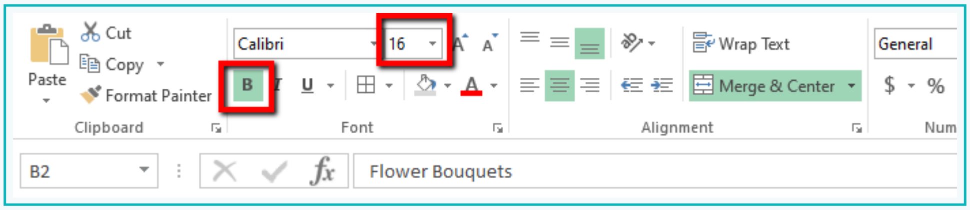
4		Student Name	Red Rose	Pink Rose	White rose	Total Roses	
5		Priyanka	5	6	4	15	
6		Rajeev	10	2	8	20	
7		Annsh	8	8	9	25	
8		Navneet	4	3	3	10	
9		Mayuri	1	3	6	10	
10			Total Bouquet			**16**	
11							

Activity

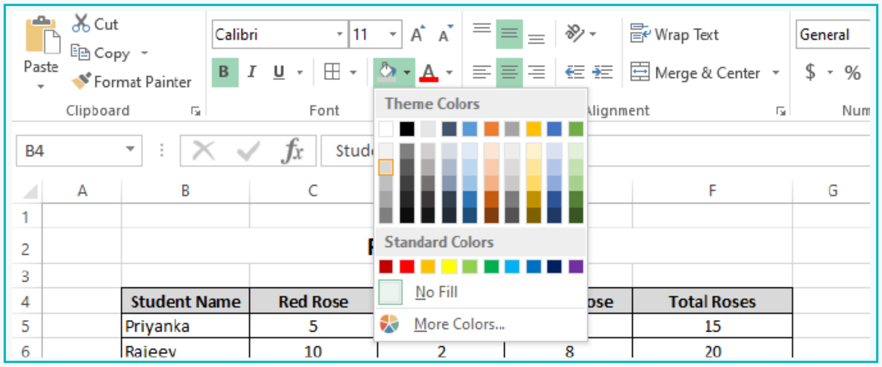
9 Select all the cells which have data and add a border to all of them together. The border separates the table from the rest of the sheet.



10 To add a title to the table, select cells from **B10 to E10** and click on **Merge & Centre.** Type **Flower Bouquets** in Bold in font size **16**.

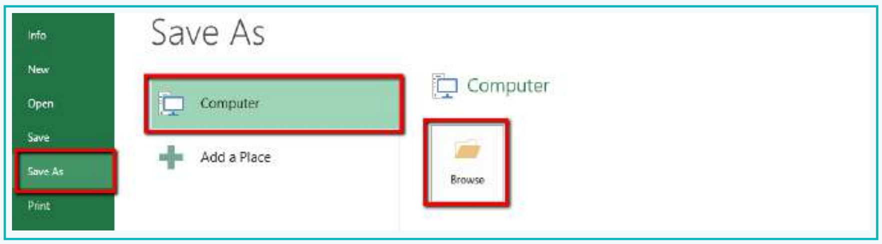


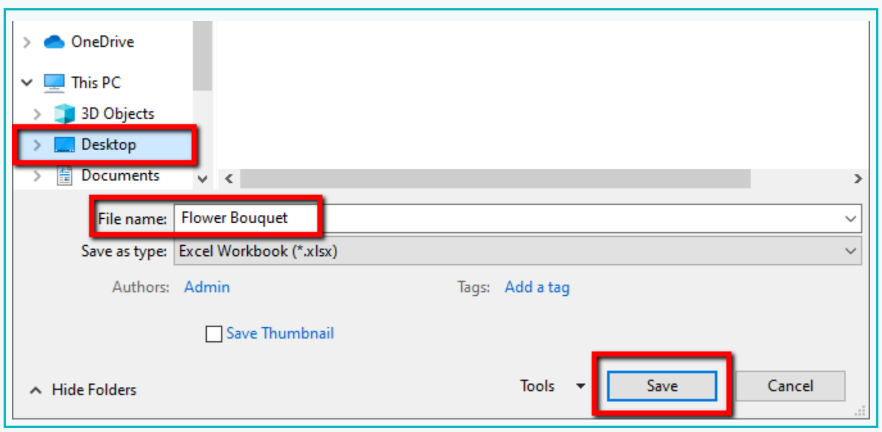
11 Change the background colour of the category cells as shown below.



Activity

12 Save the file with the name Flower Bouquet.





Final Output

Flower Bouquets

Student Name	Red Rose	Pink Rose	White Rose	Total Roses
Priyanka	5	6	4	15
Rajeev	10	2	8	20
Annsh	8	8	9	25
Navneet	4	3	3	10
Mayuri	1	3	6	10
Total Bouquet				**16**

We have learnt how to apply formulas and functions in Excel to calculate and analyse large sets of data. In this lesson, we practised the addition or sum formula: ('+', '/', SUM).

14. Spreadsheet Formulas

Tech Flash

- **Column** : Columns are the cells arranged vertically (top to bottom).
- **Row** : Rows are the cells arranged horizontally (left to right).
- **Table** : A table is a representation of data in cells, arranged in rows and columns.
- **Cell** : A cell is an intersection of a row and a column.
- **List** : A number of words or names written sequentially, typically one below the other is called a List.
- **Spreadsheet** : It is a set of rows and columns where data can be stored.